

1. The first step is to identify the problem. In this case, the problem is that the company is not meeting its sales targets.

2. The second step is to analyze the data. This involves looking at the sales figures for each product line and identifying the areas where the company is falling short.

3. The third step is to develop a plan. This involves identifying the specific actions that need to be taken to improve sales, such as increasing marketing efforts or improving customer service.

4. The fourth step is to implement the plan. This involves putting the plan into action and monitoring the results.

5. The fifth step is to evaluate the results. This involves comparing the actual sales figures to the targets and identifying any areas where further improvement is needed.

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Sixth Semester of B. Tech. (CL) Examination

May 2013

CL 310 Irrigation & Hydraulic Structures

Date: 04.05.2013, Saturday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures with pencil only wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q – 1(a) Explain: Irrigation. Justify the need of irrigation in India and describe its role in the development of the country. [04]

(b) Enlist various surface methods of irrigation and explain any ONE of them in detail. [03]

Q – 2(a) Explain: Field capacity, Frequency of irrigation. [05]

The field capacity and permanent wilting point for the given soil are 35 and 15 per cent, respectively. Determine the storage capacity of the soil within root zone which may be taken as 80 cm. At a given time the soil moisture in the field is 20 per cent and farmer applies 35 cm of water. What part of this water would be wasted? Take soil porosity = 40 % and relative density = 2.65

(b) For a irrigation project, compute the reservoir capacity with the following data: [06]

C. C. A = 40,000 ha. , Canal Losses = 13% , Time Factor = 0.70,

Reservoir Losses = 10%, Dead storage = 100 ha-m.

Crop	Intensity of Irrigation (%)	Duty (ha/cumecs)	Base period (days)
Paddy	35	850	120
Sugarcane	20	750	365
Tobacco	30	900	200
Wheat	40	1150	110

(c) Enlist different types of canal escapes and discuss their importance in canal regulation. [03]

OR

Q – 2(a) 1. Distinguish between storage efficiency and application efficiency. What are the factors affecting the application efficiency? [03]

2. The discharge at an outlet is $0.20 \text{ m}^3/\text{s}$. Average losses from outlet to the field are 10% of water flowing through the outlet. If Kor period and Kor depth for a crop are 3 weeks and 120 mm, calculate the maximum area that can be cultivated with that crop. [03]

(b) Define “Duty” and “Delta”. Also, obtain relationship between duty and delta. Suggest the measures for improving duty. [05]

(c) Discuss the functions served by a cross regulator and distributory head regulator. [03]

Q – 3(a) 1. Explain creep length and weighted creep length. Also, discuss Bligh's creep theory for the design of impervious floor on permeable soils. [04]

2. How does seepage endanger the safety of structure founded on pervious soils? [04]
What measures do you suggest for its safety?

(b) What is Exit gradient? If the depth of downstream sheet pile of a weir structure founded on fine sand is 6 m, what should be the minimum length of the impervious concrete floor required for a seepage head of 5 m? Take safe exit gradient = $1/7$ [04]

(c) Explain the difference between aqueduct and siphon aqueduct. [02]

OR

Q – 3(a) Explain, clearly, the following terms: [04]
(i) Barrage (ii) Fish ladder (iii) Weir (iv) Scouring sluices

(b) Calculate the residual uplift pressure heads and the thicknesses at the key points for the weir profile shown in Figure 1, using Bligh's theory. Also, check the safety of the weir (i) against piping and (ii) against uplift at the toe of d/s glacis, if thickness provided is 1.8 m. Take the relative density of the concrete $S = 2.4$ and coefficient of creep $C = 7$. Consider the case of no flow at pond level. [08]

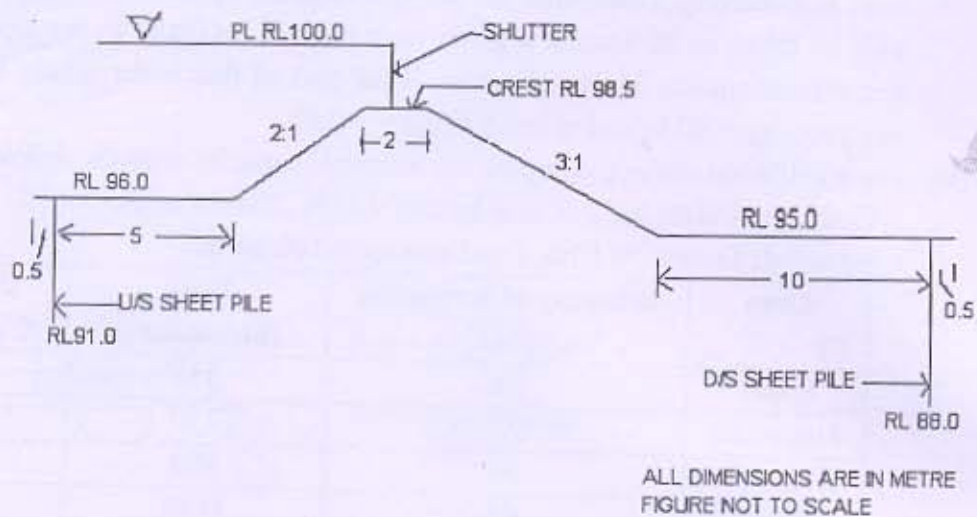


Figure 1

(c) Sketch the section of a siphon aqueduct along the drain. [2]

SECTION – II

Q – 4(a) Attempt the following: [05]

1. The forces which are considered for the analysis of an elementary profile of a gravity dam under empty reservoir conditions, are
(i) water pressure, (ii) self weight, (iii) uplift pressure, (iv) pressure due to earthquake

2. The major resisting force in a gravity dam is

(i) water pressure, (ii) wave pressure, (iii) self weight of dam, (iv) uplift pressure

3. When the reservoir is full, the maximum compressive force in a gravity dam is produced
 - (i) at the toe, (ii) at the heel, (iii) within the middle third of base, (iv) at centre of base
 4. The maximum permissible eccentricity for no tension at the base of gravity dam is
 - (i) $B/2$, (ii) $B/4$, (iii) $B/3$, (iv) $B/6$
 5. Seepage through foundation in an earthen dam is controlled by providing
 - (i) rock toe, (ii) horizontal blanket, (iii) impervious cut-off, (iv) chimney drain
 - (b) Draw a neat sketch of cross-section of rockfill dam. [02]
- Q - 5 (a) Enumerate and explain by neat sketches the different ways by which the earthen dams may fail. [04]
- (b) Check the stability of gravity dam section as shown in **Figure 2** against overturning only for reservoir full conditions. Neglect seismic forces. Take specific weight of concrete = 2.4 t/m^3 , unit weight of water = 1 t/m^3 . [05]

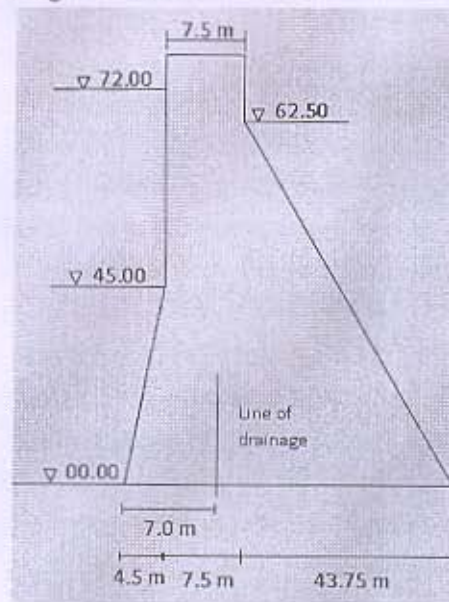


Figure 2

- (c) Draw neat sketches of following showing all the components: [05]
1. Ogee spillway with vertical & inclined upstream face
 2. Side channel spillway
- OR**
- Q - 5 (a) A gravity dam may considered safe against overturning if criteria of no tension at any point in the dam satisfied: Justify this statement. [04]
- (b) Design a concrete gravity dam for the following data: [05]
- Maximum allowable compressive strength of concrete = 3300 kN/m^2
 Maximum reservoir level = 190.0 m
 R.L. of bottom of the dam = 100.0 m
 Specific weight of concrete = 2.4
 Unit wt. of water = 10 kN/m^3
- (c) Draw neat sketches of following showing all the components: [05]
1. Straight drop spillway
 2. Chute spillway

Q - 6 (a) Design an irrigation channel to carry 500 cumecs of discharge. The channel is to be laid at a slope of 1 in 4000. The critical velocity ratio for the soil is 1.1. Use Kutter's rugosity coefficient as 0.023 and Side slope = $\frac{1}{2} : 1$. [07]

(b) Discuss briefly the various types of energy dissipators that are used for energy dissipation below overflow spillways, under different relative positions of T.W.C and J.H.C. [07]

OR

Q - 6 (a) Design a Lacey's regime channel, flowing through material of average diameter 0.328 mm, for the following data: [07]

Culturable commanded area = 1,00,000 hectare

Side slope = $\frac{1}{2} : 1$.

Intensity of irrigation:

Rabi = 40%

Kharif = 30%

Outlet discharge factor:

Rabi = 1,800 hec/cumec

Kharif = 800 hec/cumec

(b) Write short notes on:

1. Corbel
2. Cavitation

[07]

Equations:

$$C = \frac{23 + \frac{0.00155}{S} + \frac{1}{N}}{1 + \left(23 + \frac{0.00155}{S}\right) \times \frac{N}{\sqrt{R}}}$$

$$S = \frac{f^{3/2}}{4980 R^{1/3}}$$

$$S = \frac{f^{5/3}}{3340 Q^{1/6}}$$

$$C = \frac{87}{1 + \frac{K}{\sqrt{R}}}$$

$$B_2 = \sqrt{\frac{\gamma_w H_2^3}{f} \left[1 + \frac{\gamma_w^2 H_2^3}{4W_2^2} \right]}$$

$$\frac{\gamma_w S_s y_1}{24} [3B_1^2 - B_2^2 + 6X_2(B_1 + B_2) + 2B_1B_2] - \frac{\gamma_w X_2}{12} [H_1 + H_2] [2B_2 - 3X_2] - W_1 \left[\frac{B_2 - B_1}{3} - X_2 \right] = 0$$